

Lecture 1

Introduction to Digital VLSI System-on-Chip (SoC) Design

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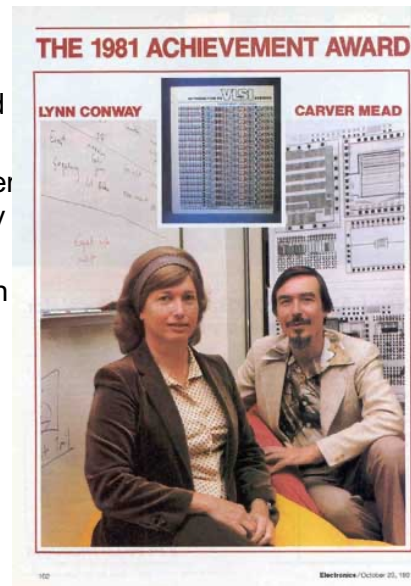
URL: www.ee.ic.ac.uk/pcheung/teaching/EE4-VLSI/

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This module is about how to turn high level specification of a fairly complex digital design into silicon layout. The CAD tool we will use is by Cadence and the targeted silicon process is TSMC 65nm low power.

Why this module? (1)

- ◆ Mead & Conway published a book in 1979 – showed students can design their own chips
- ◆ Between 1992 – 2010, I taught an elective called “Introduction to Digital IC Design”
- ◆ From 2010 onwards, Tim Constandinou took over to teach “Full-Custom IC Design” which is mainly analogue circuits
- ◆ VLSI chips are now much larger and denser than in 2011
- ◆ Design methodology is now very different



Designing IC at universities and have them fabricated and tested started with Carver Mead from Caltech and Lynn Conway in 1979. They published the first book on VLSI design that sparked interests by many universities to teach such a course in their UG programme.

EEE Department taught a similar course (by me) from 1992 to 2010. From 2010 onwards, Tim Constandinou taught a similar course, but focus on analogue IC design, with a small amount of digital.

Why this module? (2)

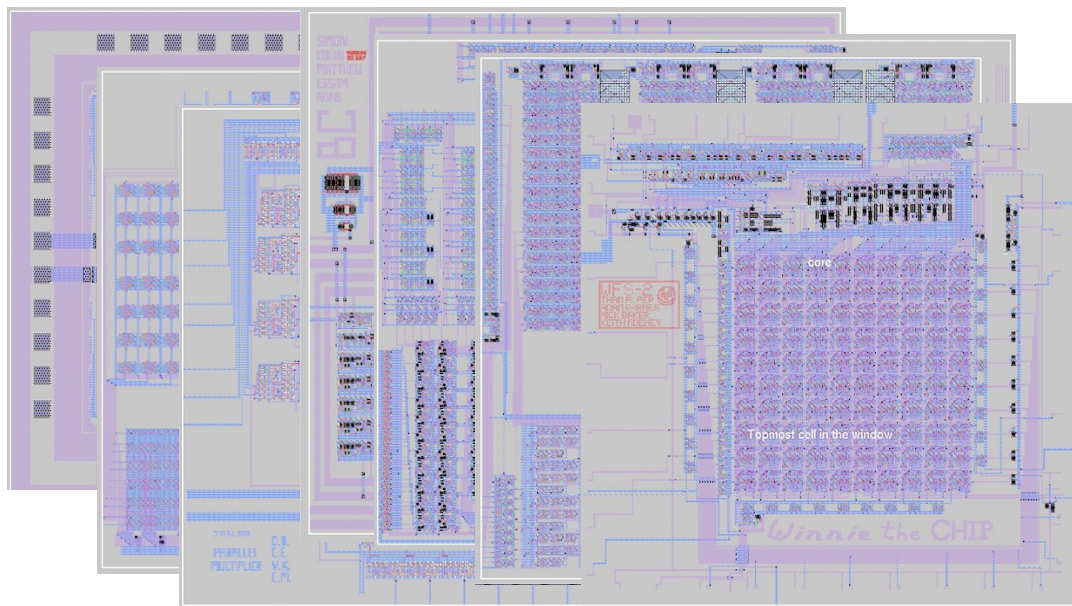
- ◆ Used to make students chips in early days
- ◆ Stopped due to high cost and chips took too long to come back
- ◆ Most were NOT tested

- ◆ Apple Inc. approached us to deliver a digital VLSI module – from specification to silicon tapeout
- ◆ They provide a grant to make chips for students again
- ◆ They help to deliver some of the industrial-related teaching

With encouragement and support from Apple Inc., we are offering a modernized version of the old VLSI course from this year onward.

With a grant from Apple, we will be able to make some of the students' design again!

Examples of students' chip layout



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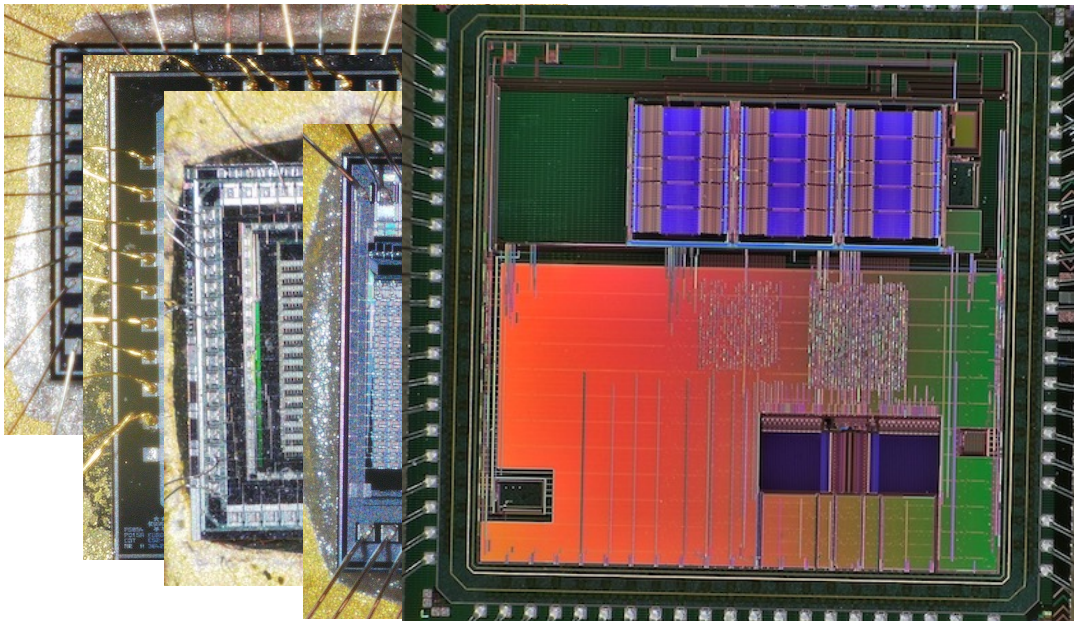
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Here are some of the layout of chips designed by EEE students who took my course. Many of them became chip designers, some in the UK and some in Silicon Valley.

These chips were all design with open-source software (mostly Magic from Berkeley for layout and IRSIM for simulation). These are relatively small chips with hundred to thousands of transistors. Therefore every polygon that make up the layout of transistors were crafted by hand!

This is of course not the appropriate way of design VLSI chip nowadays. This module is therefore completely redesigned from the ground up using industrial grade software and fabrication process.

Examples of fabricated students' designs



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Here are the fabricated chips designed by students (layout from previous slide).

The large chip is partly designed by me (and a colleague who long left academia). It was designed with LSI Logic for BBC in the late 90s.

Teaching Team



Peter Cheung
(Module lead)



Christos Bouganis
(2nd marker)



Peter Liu
(GTA)



Luca Ridolfi
(GTA)

Here is the teaching team.

Prof Christos Bouganis is my second marker and assist me where he can.

Peter Liu is a PhD student who is one of our internal experts on Cadence tools although his main interest is in analogue circuits (particularly using floating gates). He will help as a GTA.

Luca Ridolfi was our MSc student last year and he did an MSc project using Cadence tool to design a Tiny RISC-V processor. He is staying on as a GTA for our module.

Learning Outcomes

By the end of this module, students will be able to:

- ◆ Describe the **digital VLSI design flow**
- ◆ Explain the **trade-offs between the different IC design approaches**
- ◆ Understand key concepts in the **fabrication process** and the role of **process design kits (PDKs)**
- ◆ Produce **cell-based circuit** from a system-level specification
- ◆ **Design and simulate** a moderately complex digital IC
- ◆ Understand circuits from **layout**
- ◆ Perform **logical and physical verification** of the design using DRC, LVS, and other tools
- ◆ **Design for** easy post-manufacturing **testing**
- ◆ Generate a complete **tapeout deck** ready **for silicon fabrication**.
- ◆ Test that the **fabricated chip is working**

Here is a list of learning outcomes.

There may be other learning outcomes to be added when the course evolve throughout the term.

Course Design - Back-to-front

Capstone Project – Design in teams of 4 a working and easy-to-test chip

Lab experiments to teaching essential knowledge and skills

Lectures to explain underlying theories

As with EE2 CAS and EIE2 IAC, I designed this module back to front.

This is how it works: I first came up with the final project (the pinnacle of the challenges), then work out what Lab Experiments are needed for you to acquire the knowledge and skills to do the project. The Lab Experiments are designed to be self-contained.

I then plan the lectures around the Lab Experiments to ensure that you understand the underlying theories.

In this way, I make sure that the coursework and lab experiments are not only relevant, but they are synchronized to the lectures.

This approach has proved to work very well. I am teaching 5 modules this year, all based around this approach.

Assessment & Textbook

- ◆ Mid-term Lab Oral (Labs 1 to 4) (10%) – 6 November 2025
- ◆ Team Chip Design Project (40%)
- ◆ Written examination in Summer Term (40%)
- ◆ Summer Term chip test (10%)

- ◆ Textbook:
 “CMOS VLSI Design: A Circuits and Systems Perspective”, 4th Edition, ET Weste & David Harris, Pearson, 2015

Here are the assessments you will be doing. I am afraid this module is not going to be easy.

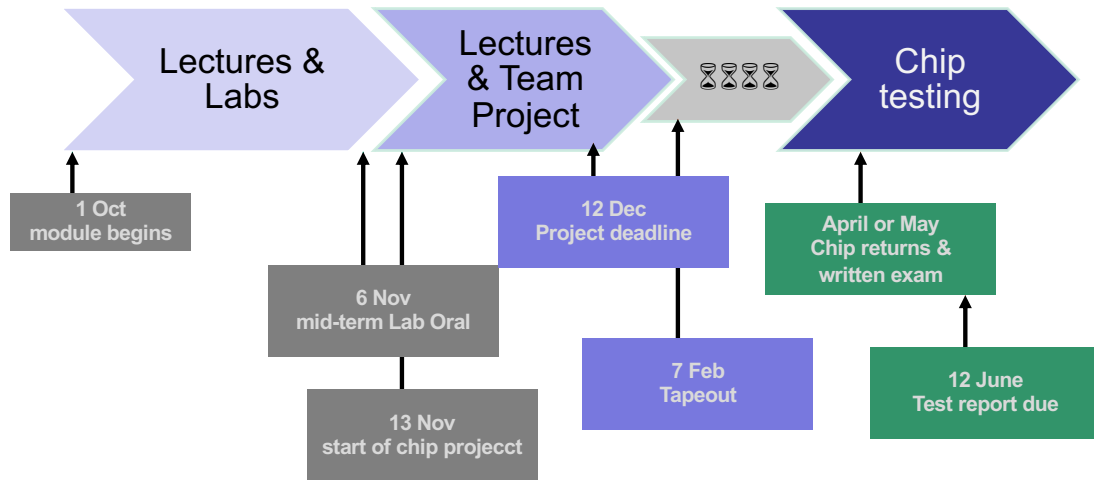
Preliminary Course schedule (2025)

Week	Lectures	Labs
1	1. Introduction and Overview 2. Design methodologies and design flow	Lab 1: Quick start with Cadence
2	3. Fabrication Process 4. CMOS Inverter & Layout Design	Lab 2: Understanding Layout (Part 1)
3	5. Full-custom combinational circuits	Lab 2: Understanding Layout (Part 2)
4	6. Full-custom sequential circuits	Lab 3: Memory and standard cells
5	7. Array Subsystems	Lab 4: Memory Self-test
6	Chip Desing Project Briefing	Mid-term Lab Oral Examination
7	8. Chip Testing 9. Design for Testability	Lab 5: Scan Insertion, LEC and Power Estimation
8	Guest Lecture on the <u>Tapeout</u> Process	Team Project Session 1
9	10. VLSI Delays 11. VLSI Interconnects	Team Project Session 2
10	12. Reliability & Variations in VLSI 13. Packaging, Power & Clock Distribution	Team Project Session 3
4 th Jan 2026	Team Project due at 23.59 GMT	

Here is a preliminary schedule for the lectures and the labs. Note the following:

- This is the first year I deliver this module. Therefore things may change over the term.
- Mid-term week has no formal teaching, but it is the week that we will launch the project.
- The deliverable is a design to be submitted via Github on the very last day of the term.
- Not included in this schedule is a small amount of work in the summer term when you have to test your fabricated chip.
- If your team's chip is not fabricated, you still have to test someone else fabricated chip according to their testing regime.

Key Dates



Here are some key dates and deadlines for this module.

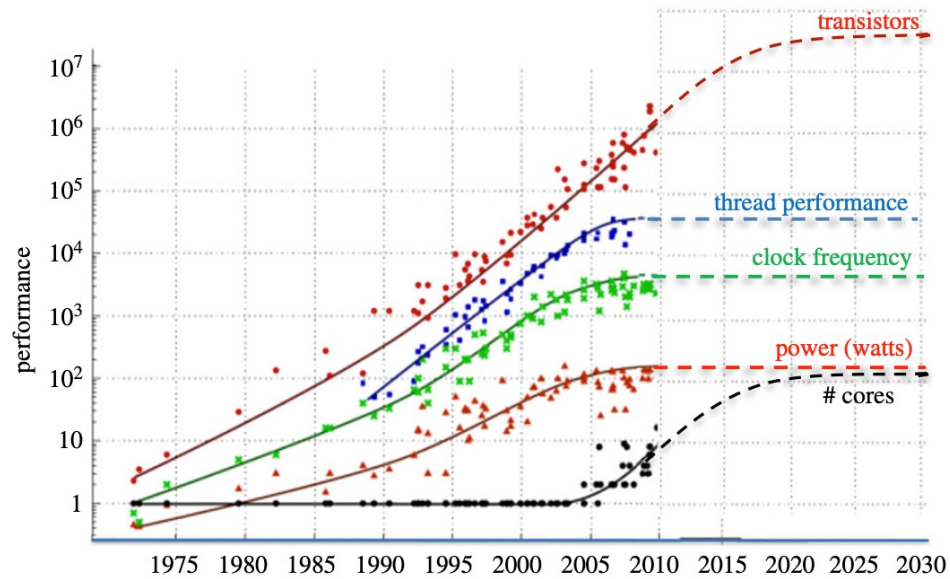
What you **MUST** do before tomorrow's Lab

- ◆ Complete and sign the two TSMC NDA forms
- ◆ Return forms to me immediately
- ◆ Fix up with ONE lab partner
- ◆ Complete the partner declaration survey

- ◆ DON'T FORGET – you cannot back out from this module!!!!
- ◆ DON'T FORGET – you **MUST** abide by the NDA agreement

YOU CANNOT START YOUR LAB UNTIL YOU HAVE SIGNED THE NDA!

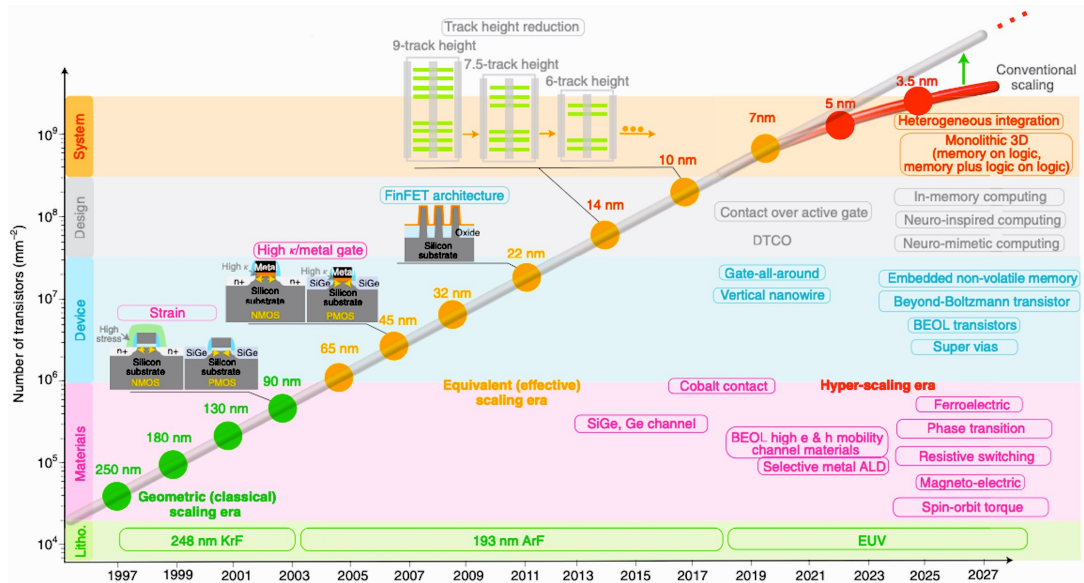
The end of Moore's Law?



Source: John Shalf (2020), "The future of computing beyond Moore's Law"

Note

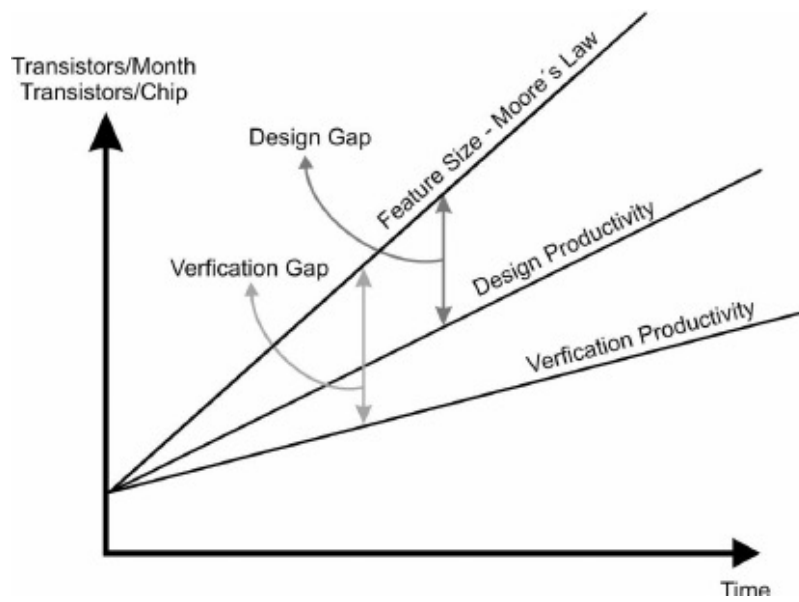
Process Technology Scaling



Source: Salahuddin, Ni and Datta, Nature electronics 2018

Note

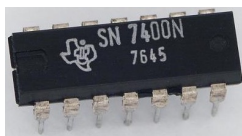
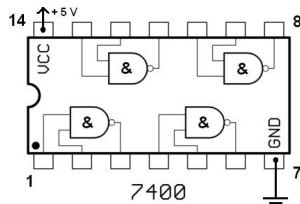
Tackling the Complexity Challenge



Source: Oswaldo Cadenas (2006)

Note

Old ways of implementing digital circuits

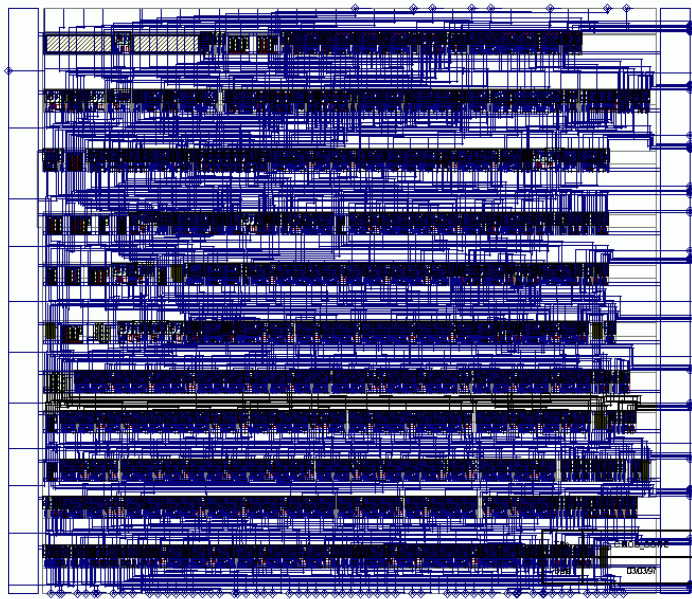


- ◆ Discrete logic – based on gates or small packages containing small digital building blocks (at most a 1-bit adder)
- ◆ De Morgan's theorem – theoretically we only need 2-input NAND or NOR gates to build anything
- ◆ Tedious, expensive, slow, prone to wiring errors

You would have been taught at least how to implement digital circuits using gates such as the one shown here. You can still buy this chip with FOUR NAND gates in one package and this is known as discrete logic. We generally **do not** use these any more. It is slow, expensive, consumes lots of energy and very hard to use.

Nevertheless, it is good to learn about NAND and NOR gates because, using De Morgan's theorem, you could in theory design and implement a Pentium microprocessor using use two input NAND or NOR gates alone. It is therefore could be regarded as the building block of all digital circuits. Similarly, you could in theory build a car using only basic Lego blocks. Unfortunately, such a car would not be very good.

Early integrated circuits based on gate arrays

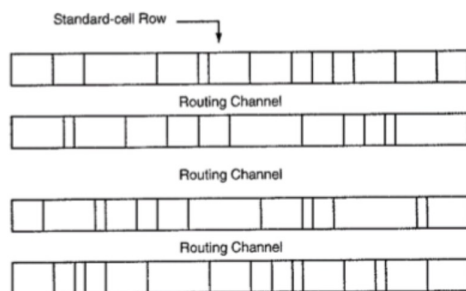
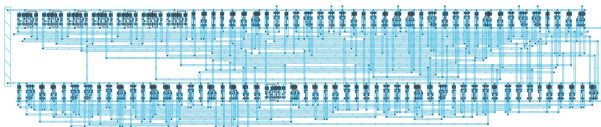
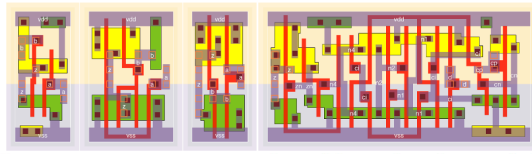


- ◆ Rows of gates – often identical in structure
- ◆ Connected to form customer specific circuits
- ◆ Can be full-custom (i.e. completely fabricated from scratch for a given design)
- ◆ Can be semi-custom (i.e. customisation on the metal layers only)
- ◆ Once made, design is fixed

Source: LSI Logic

In early days of integrated circuits, designers started using rows of basic gates (shown as the dark stuff here arranged in rows). These are either completely customised (full-custom) or it is made with standard rows of gates but leaving the gates unconnected. For a specific design, the gates are connect through wires in the wiring channels. Therefore the customisation is only in the wiring metal layers and not the layers with transistors. This is known as “semi-custom” application-specific integrated circuits (ASICs).

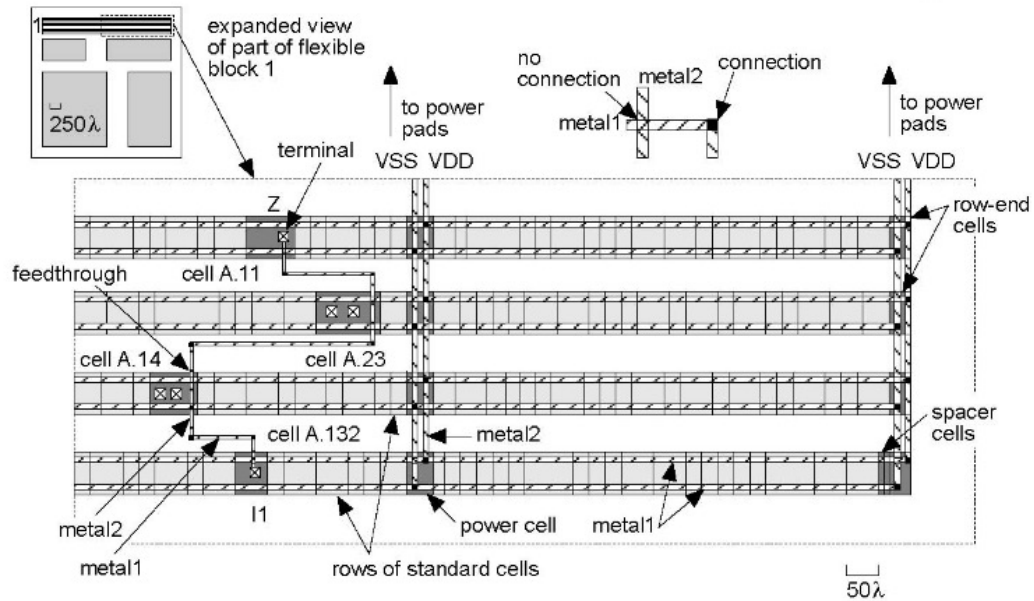
Standard cell-based design (1)



- ◆ Based on a library of logic gates and flip-flops
- ◆ Connected to form customer specific circuits
- ◆ Pitch matched – same height on all cells
- ◆ Any gaps are filled with filler cells (bridging VDD and GND)
- ◆ Bypass cells

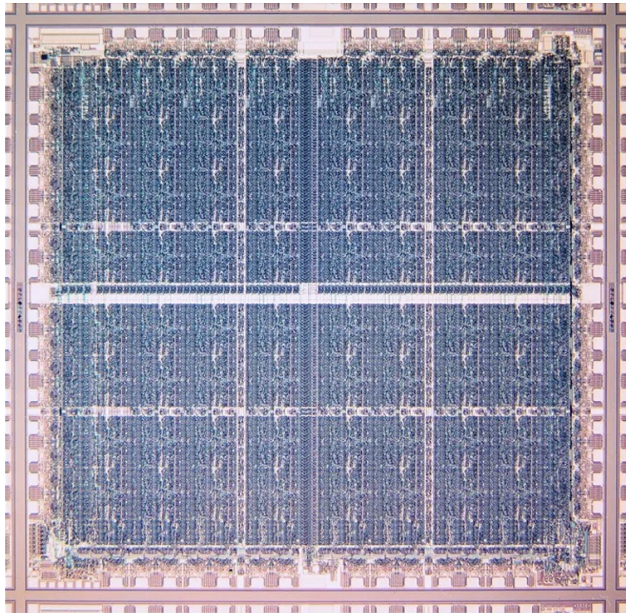
Source: VLSI Technology

Standard cell-based design (2)



FPGAs

- ◆ XC2064 die (1985)
- ◆ 2 μ m technology
- ◆ 64 CLBs (~64k trans)
- ◆ Latest AMD/Altera FPGAs have 30-40 billion trans.
- ◆ Area cost around 10x to 30x higher than ASIC
- ◆ Power may be 10x higher
- ◆ Suitable for <~100k units

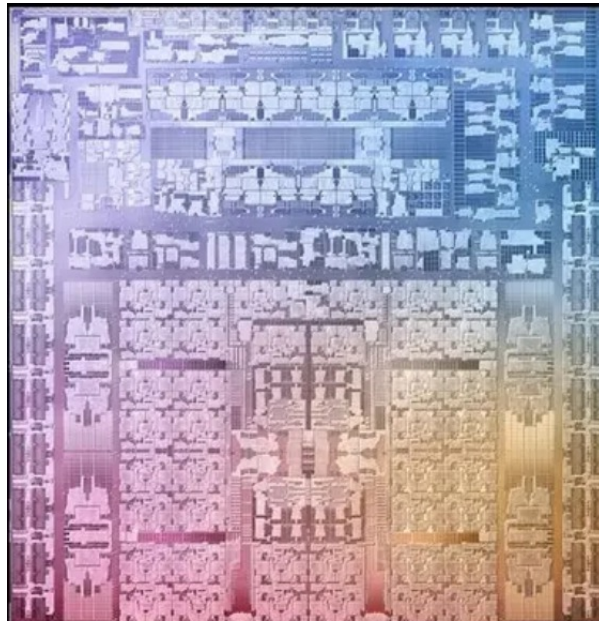


Source: Xilinx/IEEE Spectrum 2017

Note

Modern digital design – full custom IC

- ◆ Apple M3 Max
- ◆ 92 billion trans.
- ◆ 3 nm TSMC technology
- ◆ Very expensive to design
- ◆ Very expensive to manufacture
- ◆ Not viable unless >~100k units



Source: Tom's Hardware 2023

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Typical Mask Set Costs by Node

28 nm: mask set cost was around **US\$1–2 million**.

7 nm: about **US\$10–15 million**.

5 nm: about **US\$25–30 million**.

3 nm (N3 / N3B): estimated around **US\$40–50 million** for a full mask set. ([semianalysis](https://www.semianalysis.com), [asml.com](https://www.asml.com), industry analysts)

Here are some reported numbers:

A 3 nm wafer from TSMC is reported to cost around **US\$18,000**.

Sometimes it's reported a bit higher (e.g. ~\$20,000), depending on yield, contract, etc.

For comparison, 28-nm wafers reportedly used to cost ~\$5,000.

Chip sizes

Transistor Count

M3 (base)

25 billion

M3 Pro

37 billion

M3 Max

92 billion

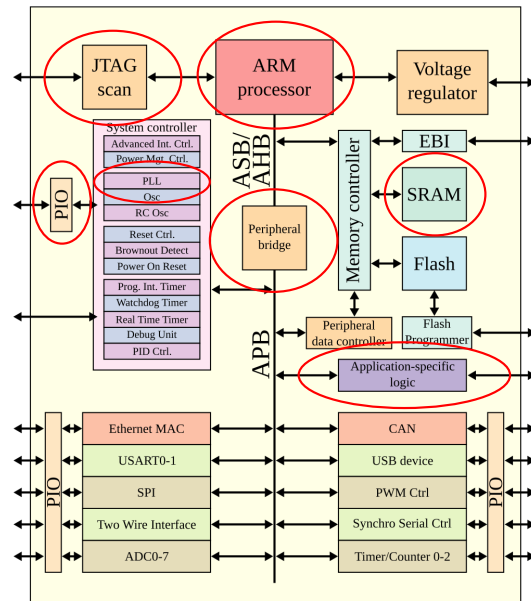
M3 Ultra

184 billion

Some reports suggest it can even push towards **US\$60 million** depending on design complexity and whether multiple variants / revisions are needed.

A Typical System-on-Chip (SoC) Design

- ◆ Here is a typical SoC
- ◆ You chip could have:
 - ARM or RISC-V processor
 - Memory
 - PIO
 - Application specific logic
 - Bus bridge
 - PLL
 - JTAG scan
 -



Source: wikipedia

Note

Comparison of VLSI Implementations

	Full Custom	Standard Cell	Gate Array	FPGA
Performance	V. High	High	Medium	Low
Density	V. High	High	Medium	Low
Development Cost	V. High	Medium	Medium	Low
Part cost	V. Low	Medium	Medium	High
Design Time	V. Long	Medium	Short	V. Short
Testability	Difficult	Less difficult	Easy	V. Easy
Application volume	High volume	Medium volume	Low	Low to V. Low

Note

Further Reading

- ◆ You may want to read the textbook Weste & Harris, sections 1.1, 1.2, 1.6, 1.8, 1.9.
- ◆ This document outlines the roadmap for VLSI in the future:
International Roadmap for Devices and Systems, 2023
https://irds.ieee.org/images/files/pdf/2023/2023IRDS_Perspectives.pdf

Here are some further readings that you may want to explore further. Of particular interest may be the International Roadmap for Devices and Systems. This document outlines the current state of the art in electronic devices and systems (not just for VLSI chips) and future challenges. It is renewed every two or three years and is used by industry and academia to guide their R&D effort.